PROTOCOLAI · AGENTVERSE UCL HACKATHON

ProtocolAI — One-Pager

ProtocolAI reviews a draft clinical trial protocol before it is locked and turns each problem into a cited suggestion a doctor approves or dismisses. Built at the AgentVerse UCL hackathon, focused on Type 2 Diabetes.

The problem

76% of protocols need at least one substantial amendment, and 23% of amendments are avoidable (Getz et al., 2024). A Phase III amendment costs a median $535,000 (Getz et al., 2016) and adds about 30 days (Tufts CSDD, 2021); industry spends $7–8 billion a year on them (Getz, 2022). By our model, every 1% of amendments prevented saves about 2.3 days and $93,000 per drug programme.

Who it helps

Pharma and biotech sponsors and the CROs that design and run their trials (industry started 2,248 Phase 2–3 trials in 2024), and the clinicians who sign protocols off, for whom each finding is a Decision Card to approve or dismiss.

Our data

- **The draft protocol:** parsed into 11 ICH M11 facets; every fact keeps its quote and page.
- **ClinicalTrials.gov, searched live:** peer trials with the same phase and age group; values outside the peers' 10th–90th percentile are flagged.
- **283 guideline rules:** ICH M11, SPIRIT 2025, EMA and FDA guidance, ICH E9(R1) and FDAAA.
- **CDC NHANES:** eligibility rules run on real patients with diabetes to show which criteria exclude the most.

How we evaluated it

We ran it on the original PIONEER 6 protocol (Novo Nordisk, oral semaglutide, version 3.0, February 2016) and compared its output with the amendments made before enrolment. It flagged 3 of the 4 major changes, each with a fix.

REAL AMENDMENT	PROTOCOLAI ON THE DRAFT	RESULT
Retinopathy exclusion with eye exam within 90 days (Jul 2016)	“Exclude proliferative or severe retinopathy needing treatment; require fundoscopy within 90 days”	**Matched**
Eye exams during treatment (Nov 2016)	“Schedule eye examinations and UACR measurements”	**Matched**
Highly effective contraception, Germany (Feb 2016)	“Define childbearing potential, list methods and give the duration”	**Matched**
Bicarbonate test, SGLT2 background drugs (Nov 2016)	No finding	**Missed**

This is one trial and one agent. Next we measure extraction accuracy, historical hit-rate and clinician acceptance across many protocols.

Limits and safeguards

Limits. It can only catch avoidable amendments. Some rules come from guidance newer than older protocols, so backtests can overstate foresight. Savings are modelled, not measured. Drug-class rules don't yet cover background medicines, and NHANES can't model every criterion or predict site recruitment.

Safeguards. Nothing changes without physician sign-off, and safety thresholds are never relaxed automatically. Every quote is checked against its page, what can't be verified is labelled rather than guessed, every flag gets a second opinion, and failed steps show as partial.